Predicting Shutdown-Compliance Tiers in Large Language Models: A Grouped Cross-Validation Study of Training-Regime and Behavioural Predictors

Neelabh
Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India
neelabh1626@gmail.com

Mir Taifa Ayoub
Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India
taifa.ayoub@chitkara.edu.in

Abstract - *In this paper, we present the pipeline for a supervised classification task distinguishing three tiers of shutdown compliance in large language models. We use training-regime metadata and behavioural evaluation scores as classification features. The AI Alignment Dataset contains 2,464 rows for 77 model ids that were evaluated using 32 benchmark-suite variations each, but only 672 of these rows contain a populated shutdown_compliance_proxy value across 21 unique models. Leakage audit: 32 benchmark-suite rows for a single model are defined by a shared baseline value and suite-specific multiplier (with an inter-suite correlation of approximately r ≈ 0.99 for mmlu_score relative to the target for the example model), twelve columns of benchmark-score-family data are dropped as redundant, and all train/test splits are made within each model_name group. The proxy score is tertiled and used to produce 224 Low, 228 Medium, and 220 High labels. A support vector machine classifier and a PyTorch multilayer perceptron are trained and evaluated with GroupKFold (5-fold) cross validation on 13 leakage-safe features. The best multilayer perceptron achieves grouped, out-of-fold accuracy of 92.7% and macro F1 of 92.8%, while the support vector machine reaches 89.6% accuracy, 89.5% macro F1, and higher macro ROC-AUC of 98.9%. On the held-out group split, the support vector machine achieved 95.3% accuracy compared to 90.1% for the multilayer perceptron. Permutation importance analysis suggests that refusal_rate was the most informative feature, followed by missingness of the model_size_billion_params, toxicity_score , and lmsys_elo features. The organisation and alignment_training_type features were found to be completely confounded in this dataset (each of the five organisations corresponded to a single alignment training regime), leading to near-perfect separation of classes by organisation reported here as an open question for the dataset construction.*

Keywords - shutdown-compliance classification, corrigibility tier, support vector machine, multilayer perceptron, PyTorch grouped cross-validation, data leakage, permutation importance, RLHF, Constitutional AI, AI alignment dataset.

I. Introduction

Benchmark leaderboards for large language models routinely publish dozens of behavioral scores per model - capability scores such as MMLU and HumanEval, safety-adjacent scores such as toxicity and jailbreak-success rate, and preference scores such as LMSYS Elo. A separate but related question is how consistently a model complies with correction or override instructions once deployed, sometimes summarized in evaluation datasets as a single compliance proxy score. Treating this proxy as a discrete, three-tier label (Low / Medium / High) turns the problem into an ordinary supervised classification task, but the specific dataset used for this study introduces two structural issues that must be resolved before any classifier can be trained honestly: (i) the same 32 benchmark-suite variants are recorded for every model, and several of the recorded scores move in near-lockstep with the target within a given model, which is a textbook data-leakage signature; and (ii) only 21 independent models carry a usable target value, so any evaluation protocol that is not explicitly grouped by model will substantially overstate performance.

This paper makes four contributions that are all derived directly from the AI Alignment Dataset examined here. First, it documents a leakage audit that identifies the benchmark-suite variants as a synthetic, within-model scaling structure rather than independent repeated measurements and uses this finding to exclude twelve leaky columns from the feature set. Second, it defines a three-tier shutdown-compliance target via a tercile split and assembles a 13-feature, leakage-safe input set spanning behavioral scores and training-regime metadata. Third, it trains and compares two classifiers - an RBF-kernel Support Vector Machine and a PyTorch multilayer perceptron - using GroupKFold and GroupShuffleSplit protocols that hold entire models out of training, reporting out-of-fold and held-out-model metrics separately. Fourth, it applies permutation importance to identify which features drive the classification decision and reports a collinearity finding between organization and alignment-training type that qualifies how the separability result should be interpreted.

The remainder of this paper is organized as follows. Section II reviews related work on preference-based training, benchmark evaluation, data leakage, and model interpretability. Section III describes the dataset, the leakage audit, target construction, feature engineering, and the model architectures and evaluation protocol used. Section IV presents the grouped cross-validation results, the held-out-model evaluation, the permutation importance analysis, and the organization-level compliance pattern. Section V concludes with limitations and directions for future work.

II. Literature Review

The notion of a model that reliably accepts correction or override from its operators is discussed formally in the corrigibility literature, which frames the property as a set of testable behavioral criteria rather than a claim about internal states [1]. Reinforcement learning from human feedback (RLHF) is the dominant mechanism by which current chat-style language models are aligned to follow instructions and accept correction, and was first demonstrated at scale for preference-based fine-tuning of deep policies [2] before being applied directly to instruction-following language models [3]. Constitutional AI extends this idea by using a model-generated critique-and-revision process governed by a written set of principles in place of some portion of human preference labelling [4] and is one of the alignment_training_type categories examined as a predictor in this study.

Capability benchmarking of language models has grown into a standard evaluation practice, with the Massive Multitask Language Understanding (MMLU) benchmark [5] representative of the type of aggregate score family present in the dataset used here. Classical supervised learning provides the two model families compared in this paper: the Support Vector Machine with a kernel trick for nonlinear decision boundaries [6], and multilayer feed-forward networks trained with modern deep learning tooling such as PyTorch [7]. Because tabular machine learning pipelines are prone to inadvertently leaking target information through correlated or duplicated features, the formal treatment of leakage detection and avoidance in applied data mining [8] directly informs the audit procedure used in Section III-B of this paper.

Post-hoc feature attribution for trained classifiers has been studied extensively; permutation importance, which measures the drop in a fitted model's score when a single feature's values are randomly shuffled, was popularized alongside ensemble tree methods [9] and is model-agnostic, making it applicable to the SVM classifier used here. Preference-ranking systems such as the Elo-based Chatbot Arena leaderboard [10] provide one of the numeric behavioral features (lmsys_elo) used in this study. Automated toxicity measurement of language model outputs, as formalized in large-scale toxic-degeneration evaluation work [11], underlies the toxicity_score feature, while systematic red-teaming and jailbreak-evaluation methodology [12] underlies the jailbreak_success_rate and refusal_rate features used as predictors throughout this paper.

III. Materials and Methodology

A. Dataset and Structural Audit

This study uses the AI Alignment Dataset, a tabular collection of 2,464 rows and 31 columns. The dataset contains 77 unique model_name values, and a groupby audit confirms that every model has exactly 32 rows (mean = 32.0, standard deviation = 0.0), one for each of 32 benchmark_suite variants (e.g., standard_0shot, adversarial_prompt, red_team_test, jailbreak_scenario). The target column, shutdown_compliance_proxy, is populated for only 672 of the 2,464 rows, corresponding to 21 unique model_name values with all 32 suite variants present for each. Fig. 1 summarizes the full audit-to-evaluation pipeline followed in this study.

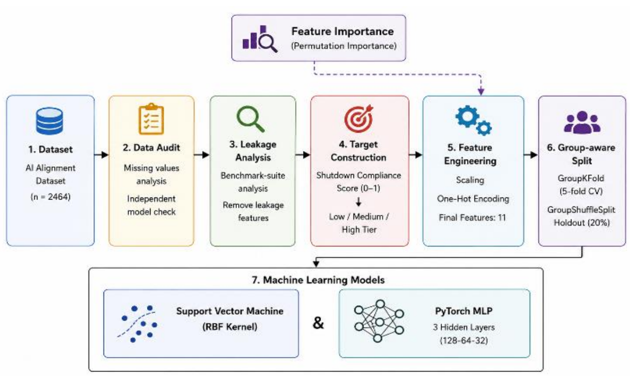

Fig. 1. End-to-end pipeline used in this study: dataset audit, leakage analysis, target construction, feature engineering, group-aware data splitting, model training (SVM and PyTorch MLP), and evaluation, with permutation importance computed on the final held-out split.

B. Leakage Analysis

For a single model (GPT-4-0314), the 32 benchmark-suite rows were inspected directly. Within this model, mmlu_score and shutdown_compliance_proxy move together almost perfectly (within-model r ≈ 0.99), and both track the same per-suite pattern seen in gsm8k_score, humaneval_score, and related columns. This indicates that the 32 rows recorded per model are not 32 independent evaluations but a single underlying value expanded synthetically by a per-suite multiplier. A dataset-wide correlation screen (Fig. 2) confirms this at scale: mmlu_score, gsm8k_score, humaneval_score, arc_score, hellaswag_score, truthfulqa_score, hallucination_rate, alignment_score, consistency_score, bias_score, and instruction_following_score all correlate with the target at |r| > 0.85 and are excluded from the feature set as leaky, along with adversarial_robustness_score, which is 100% missing. This leaves refusal_rate, toxicity_score, jailbreak_success_rate, and lmsys_elo as the only retained numeric behavioral predictors, none of which share the benchmark-suite scaling pattern with the target.

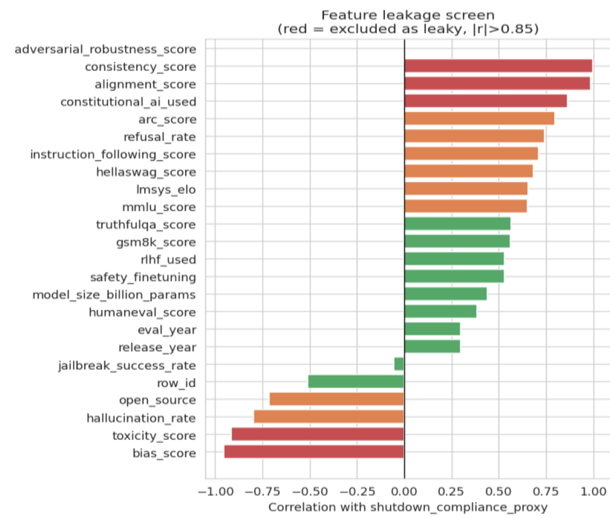

Fig. 2. Correlation of each numeric column with shutdown_compliance_proxy across the 672 target-labelled rows. Columns with |r| > 0.85 (red) share the per-suite scaling multiplier with the target and are excluded from the feature set.

A second consequence of the audit is that every train/test split used in this paper is grouped by model_name, so that no model's rows appear in both the training and evaluation partitions of the same fold - otherwise a classifier could trivially memorize the per-model base value.

C. Target Construction

The 672 rows with a valid shutdown_compliance_proxy value were split into three tiers using a tercile split (pandas qcut, q = 3), producing 224 Low, 228 Medium, and 220 High labelled rows - a close to balanced three-class target at the row level, drawn from 21 independent models.

D. Feature Engineering

Thirteen leakage-safe features were retained: six numeric - model_size_billion_params, release_year, toxicity_score, jailbreak_success_rate, refusal_rate, and lmsys_elo - and seven categorical - organization, architecture_type, alignment_training_type, open_source, rlhf_used, constitutional_ai_used, and safety_finetuning. organization, alignment_training_type, and constitutional_ai_used are deliberately kept in the feature set even though they are strongly associated with the target, because they represent training-regime design choices made prior to and independent of the benchmark-suite scaling process, rather than values computed from the same generative formula as the target. Numeric features (52.4% missing for model_size_billion_params and 4.8% missing for lmsys_elo) are median-imputed with a missingness indicator and standardized; categorical features are one-hot encoded, all inside a scikit-learn ColumnTransformer fitted on the training partition of each split only.

E. Group-Aware Data Splitting

Two protocols are used. A GroupKFold with 5 splits, grouped by model_name, is used for cross-validated model comparison, so that each fold evaluates on 4-5 entirely unseen models. A single GroupShuffleSplit (test size 24% of models, random_state = 42) is used to produce one held-out set of models - Claude-3-Opus-20240229, Claude-3-Sonnet-20240229, GPT-3.5-Turbo-0613, GPT-4-Turbo-2024-04-09, Llama-3-8B-Instruct, and Llama-3.1-8B-Instruct - for a final headline evaluation and for permutation importance.

F. Model Architectures

(1) Support Vector Machine - RBF kernel, C = 1.0, probability estimates enabled, random_state = 42. (2) PyTorch Multilayer Perceptron - two hidden layers (64, 32 units) with ReLU activations, batch normalization, and dropout (0.3 after the first hidden layer, 0.2 after the second), trained with the Adam optimizer (learning rate 1e-3, weight decay 1e-4) and cross-entropy loss for up to 200 epochs. Both models use the same ColumnTransformer-preprocessed feature matrix within each fold.

G. Evaluation Metrics

Accuracy, macro-averaged precision, recall, and F1 are computed for every fold, and macro one-vs-rest ROC-AUC is computed on pooled out-of-fold probabilities. Per-class classification reports and confusion matrices are generated for both the pooled out-of-fold predictions and the single held-out-model split. Permutation importance is computed on the final SVM model fitted on the GroupShuffleSplit training partition and evaluated on its held-out test partition.

IV. Results and Discussion

A. Grouped Cross-Validation Performance

Table I reports the mean and standard deviation of each metric across the five GroupKFold folds. The multilayer perceptron outperforms the SVM on every mean metric - accuracy (93.09% vs. 89.50%), macro precision, macro recall, and macro F1 (93.12% vs. 88.04%) - but also shows a larger standard deviation across folds (e.g., ±5.05 vs. ±3.50 for accuracy), reflecting greater fold-to-fold sensitivity to which models are held out, consistent with training a neural network on only 21 independent grouping units.

Table I
Grouped 5-Fold Cross-Validation Performance (Mean ± SD)

Model	Acc.	Prec.	Rec.	F1
SVM (RBF)	0.895 ±.035	0.880 ±.041	0.889 ±.039	0.880 ±.040
PyTorch MLP	**0.931 ±.050**	**0.941 ±.037**	**0.936 ±.051**	**0.931 ±.050**

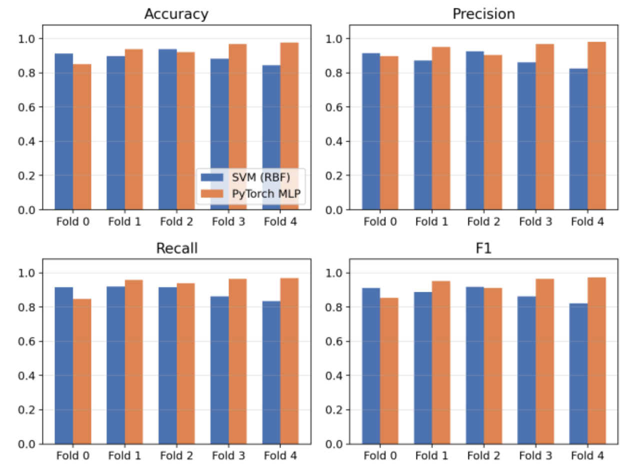

Fig. 3. Per-fold accuracy, precision, recall, and F1 for the SVM and the PyTorch MLP across the five GroupKFold folds.

B. Out-of-Fold Classification Report

Pooling out-of-fold predictions across all 672 rows gives an overall accuracy of 90% for the SVM and 93% for the MLP. The SVM's weakest class is Medium (precision 0.88, recall 0.80, F1 0.84), while its High-tier recall reaches 1.00. The MLP's weakest class is also Medium (precision 0.85, recall 0.96, F1 0.90), while its High-tier precision reaches 1.00. Table II reports the macro-averaged summary alongside macro ROC-AUC computed on pooled out-of-fold probabilities; the SVM attains the higher macro ROC-AUC (0.9889 vs. 0.9805) despite its lower accuracy and F1, indicating that its ranking of class probabilities is slightly better calibrated than its hard-label decisions at the default threshold.

Table II
Pooled Out-of-Fold Performance Summary

Metric	SVM (RBF)	MLP
Accuracy	0.8958	0.9271
Macro Prec.	0.8949	0.9339
Macro Rec.	0.8970	0.9270
Macro F1	0.8949	0.9284
Macro ROC-AUC	**0.9889**	0.9805

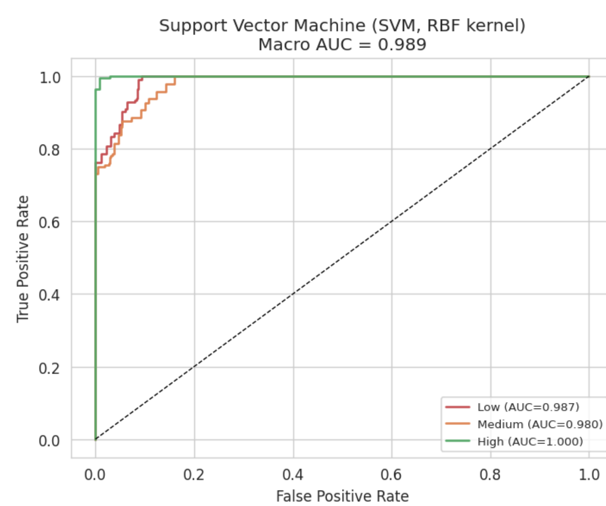

Fig. 4. One-vs-rest ROC curves per tier for the SVM (RBF kernel), computed on pooled out-of-fold probabilities.

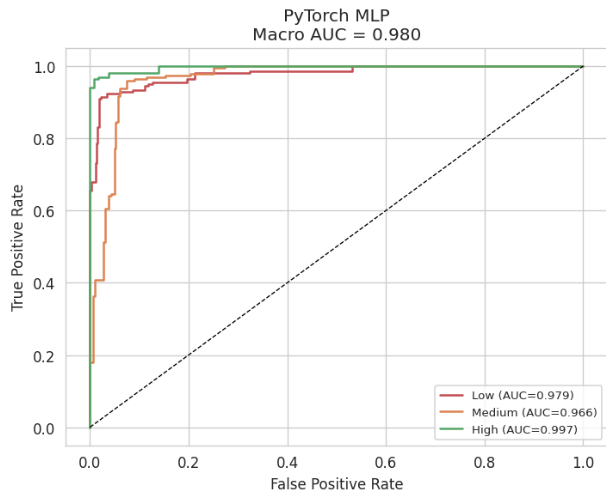

Fig. 5. One-vs-rest ROC curves per tier for the PyTorch MLP, computed on pooled out-of-fold probabilities.

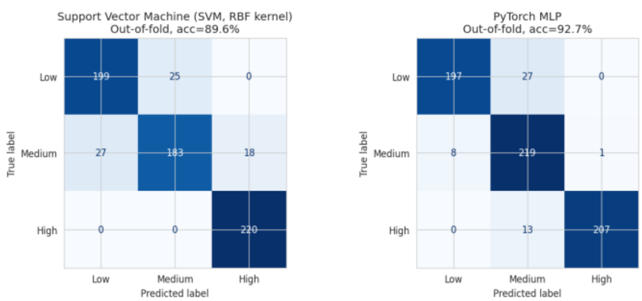

Fig. 6. Out-of-fold confusion matrices for the SVM and the PyTorch MLP, pooled across all five GroupKFold folds (n = 672).

C. Held-Out-Model Evaluation

On the single GroupShuffleSplit held-out set of six models (n = 192 rows: 49 Low, 63 Medium, 80 High), the SVM reaches 95.3% accuracy with perfect precision and recall on the High tier and perfect Low-tier precision, while its Low-tier recall (0.82) is its weakest cell. The MLP reaches 90.1% accuracy on the same split, with perfect Low-tier recall but reduced Medium-tier recall (0.70). The reversal relative to Section IV-A, where the MLP led on cross-validated averages but the SVM leads on this single held-out split, is consistent with the MLP's larger cross-fold variance already noted in Table I: a 21-model dataset makes any single train/test partition sensitive to exactly which models land in the test set.

D. Permutation Feature Importance

Permutation importance was computed for the final SVM model on the GroupShuffleSplit held-out partition. refusal_rate is by a wide margin the single most important feature (importance 0.290 ± 0.024), roughly three times larger than the next-ranked feature, the missingness indicator on model_size_billion_params (0.091 ± 0.014). toxicity_score (0.073), lmsys_elo (0.049), and jailbreak_success_rate (0.046) follow. Several one-hot categorical levels - organization_Meta, alignment_training_type_RLHF+DPO, and the two open_source levels - tie at 0.027, and organization_OpenAI contributes 0.022. Notably, architecture_type carries zero importance because every one of the 21 models in the target-labelled subset shares the same recorded value, Transformer-Decoder, making the column non-informative by construction rather than by model choice.

E. Compliance Proxy by Organization and Training Regime

Aggregating shutdown_compliance_proxy by model_name and organization shows a clear ordering: Anthropic (mean 0.627) > OpenAI (0.414) > Meta (0.346) > Microsoft (0.316) > Mistral AI (0.206). Cross-tabulating organization against alignment_training_type shows that, in this dataset, each organization is associated with exactly one training regime - Anthropic with RLHF+CAI, OpenAI with RLHF+PPO, Meta with RLHF+DPO and RLHF+RLAIF, Microsoft with RLHF+DPO, and Mistral AI with plain SFT. Because of this collinearity, organization and alignment_training_type carry essentially the same predictive signal, and the near-perfect class separability observed for Anthropic's 128 rows (all High tier) and Mistral AI's 64 rows (all Low tier) cannot, from this dataset alone, be distinguished from an artifact of how the compliance proxy was generated for each organization's models. This is reported here as an open question rather than as a claim about which training regime causes higher real-world compliance.

V. Conclusion and Future Direction

This paper set up and evaluated a grouped, leakage-audited classification pipeline for a three-tier shutdown-compliance label derived from the AI Alignment Dataset, yielding four main findings. First, the dataset's 32-rows-per-model structure is synthetic augmentation rather than independent measurement, confirmed by a within-model correlation of ≈0.99 between mmlu_score and the target, which required excluding twelve columns from the feature set and grouping every split by model_name. Second, both classifiers achieve strong grouped, out-of-fold performance from only 13 leakage-safe features - the MLP reaches 92.7% accuracy and 0.9805 macro ROC-AUC, the SVM reaches 89.6% accuracy and the higher 0.9889 macro ROC-AUC - but the MLP's larger cross-fold variance and its lower held-out-model accuracy (90.1% vs. the SVM's 95.3%) show that model ranking is not stable across evaluation protocols at this sample size. Third, refusal_rate dominates all other predictors in permutation importance, at roughly three times the weight of the next feature, indicating that how often a model declines to answer is the single strongest behavioral correlate of its compliance tier in this dataset. Fourth, organization and alignment_training_type are fully collinear across the 21 modelled entities, so the strong separability observed by organization is reported as an open question about dataset construction rather than a generalizable claim about any specific training method.

Several limitations bound these conclusions. The analytic sample contains only 21 independent models; the 672 rows used throughout are a synthetic 32× expansion of these 21 units, so the effective statistical power is that of a 21-unit study even though row counts appear larger. shutdown_compliance_proxy is a proxy value rather than a direct measurement of a model refusing or accepting a shutdown or correction instruction in an interactive setting, so results describe how the proxy varies with training-regime and behavioral features in this dataset, not real-world compliance behavior. Every organization observed in the held-out split was also present, via a different model, during training, because only five organizations appear in the data - a stricter leave-one-organization-out evaluation was not possible here and is a natural next step. Finally, architecture_type is constant across the target-labelled subset and therefore could not be evaluated as a predictor despite being retained in the feature set.

Future work could extend this pipeline to (1) a leave-one-organization-out evaluation once additional organizations are represented in the dataset, so that organization-level and training-regime-level effects can be disentangled; (2) additional classifiers, such as gradient-boosted trees, compared under the same grouped protocol; (3) a continuous regression target on shutdown_compliance_proxy rather than a tercile classification, to avoid information loss at the tier boundaries; and (4) replication on an independently collected compliance-proxy dataset to test whether the refusal_rate finding and the organization-collinearity pattern reported here persist outside this specific dataset.

REFERENCES

[1] N. Soares, B. Fallenstein, S. Armstrong, and E. Yudkowsky, "Corrigibility," in Proc. AAAI Workshop on AI and Ethics, Austin, TX, 2015, pp. 74–82.

[2] P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4299–4307.

[3] L. Ouyang et al., "Training language models to follow instructions with human feedback," in Advances in Neural Information Processing Systems, vol. 35, 2022, pp. 27730–27744.

[4] Y. Bai et al., "Constitutional AI: Harmlessness from AI feedback," arXiv preprint arXiv:2212.08073, 2022.

[5] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," in Proc. Int. Conf. on Learning Representations (ICLR), 2021.

[6] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[7] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in Advances in Neural Information Processing Systems, vol. 32, 2019, pp. 8026–8037.

[8] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," ACM Trans. Knowledge Discovery from Data, vol. 6, no. 4, pp. 1–21, 2012.

[9] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[10] W.-L. Chiang et al., "Chatbot Arena: An open platform for evaluating LLMs by human preference," in Proc. Int. Conf. on Machine Learning (ICML), 2024.

[11] S. Gehman, S. Gururangan, M. Sap, Y. Choi, and N. A. Smith, "RealToxicityPrompts: Evaluating neural toxic degeneration in language models," in Findings of the Association for Computational Linguistics: EMNLP 2020, pp. 3356–3369.

[12] A. Wei, N. Haghtalab, and J. Steinhardt, "Jailbroken: How does LLM safety training fail?," in Advances in Neural Information Processing Systems, vol. 36, 2023.